

Ph134, Lecture 3

- **Lessons from last time:**
 - **sample variance technique: Mean +/- SEM**
 - **issues to be aware of**
- **Fitting a function to a data set:**
 - **justification for χ^2 fitting --**
The principle of maximum likelihood
 - **outputs of a χ^2 fit**

From **N** trials to a value and an uncertainty:

Given N independent trials $\{y_1, y_2, \dots, y_N\}$, i.e., N values sampled at random from the distribution $P(y)$,

we conclude $\mu = \bar{y} \pm s_{\bar{y}}$, where $s_{\bar{y}} = \frac{s}{\sqrt{N}}$

and

$$s = \sqrt{\frac{1}{N-1} \sum_{n=1}^N (y_n - \bar{y})^2}$$

Issues to be careful about?

- make the trials as independent as possible! E.g., if measuring $y(x)$, reset x carefully before each trial.*
- accounting for random error in all this, but never for systematic error*

What do we mean by “best” estimate of μ ?

Making an estimate of μ means making a guess about the parameters of the true underlying distribution from which the y_n values are drawn.

If we assume a particular underlying distribution, we can calculate from it the probability of all our y_n values coming out the way they did.

The Principle of Maximum Likelihood:

The max likelihood estimate of μ for the underlying distribution is the one that maximizes the probability of the measurement results we got.

What do we mean by “best” estimate of μ ?

Making an estimate of μ means making a guess about the parameters of the true underlying distribution from which the y_n values are drawn.

If we know the underlying distribution is normal, then we can show that $\bar{y}$ is the max likelihood estimator of μ as well as being unbiased.

The Principle of Maximum Likelihood:

The max likelihood estimate of μ for the underlying distribution is the one that maximizes the probability of the measurement results we got.

We've just finished with the “sample variance” approach to estimating a single quantity through repeated trials.

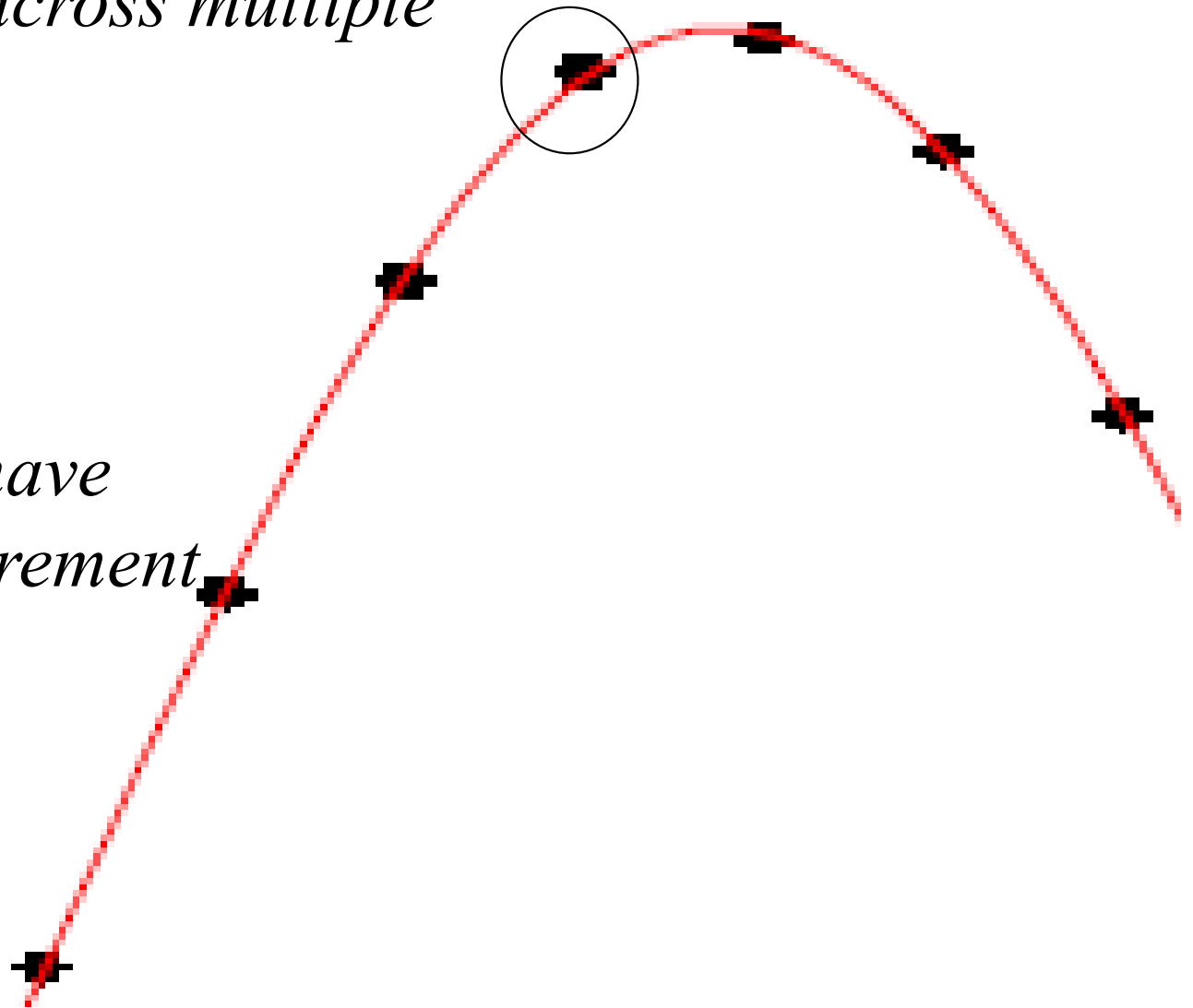
We can do this for multiple different quantities – specifically, for $y(x)$ across multiple different values of x .

This will give us

$$\{(x_m, \bar{y}_m, s_{\bar{y}_m})\} \dots$$

i.e., for each x_m we have done repeated measurement trials so we can say

$$y(x_m) = \bar{y}_m \pm s_{\bar{y}_m}$$



Fitting a function to a data set:

Given: M data points with uncertainties, $\{(x_m, \bar{y}_m, s_{\bar{y}_m})\}$,
and a proposed relationship $y = f(x; a, b, \dots)$
where $a, b, \dots$ are free parameters in the function f

Task: Find the values of the free parameters $a, b, \dots$ which
allow f to best fit the data. Calculate the uncertainties of
the parameters and find a quantitative measure of how
well the function f in fact describes the data.

What do we mean by “best” fit parameters?

Making estimates of $a, b, \dots$ means making an (informed) guess about the parameters of the true underlying distributions from which the $\bar{y}_m$ values are drawn.

If we assume particular distributions, we can calculate the probability of all our $\bar{y}_m$ values coming out the way they did.

The Principle of Maximum Likelihood:

The max likelihood estimates of parameter values for the underlying distribution are the ones that maximize the probability of the measurement results we got.

Guess that the underlying relationship is $y = f(x)$.

(E.g., suppose $f(x) = ax + b$ and guess parameter values a, b .)

*Assume each $\bar{y}_m$ comes from a normal distribution
w/ true mean $f(x_m)$ and variance $s_{\bar{y}_m}^2$*

$P(\text{data point 1 comes out w/in } \Delta y \text{ of our result})$

$$= P(\bar{y}_1) \Delta y = \frac{1}{\sqrt{2\pi s_{\bar{y}_1}^2}} e^{-\frac{[\bar{y}_1 - f(x_1)]^2}{2s_{\bar{y}_1}^2}} \Delta y$$

$P(\text{all } M \text{ data points come out w/in } \Delta y \text{ of our results})$

$$\begin{aligned} &= P(\bar{y}_1) P(\bar{y}_2) \cdots P(\bar{y}_M) (\Delta y)^M \\ &= \left(\frac{1}{\sqrt{2\pi s_{\bar{y}_1}^2}} e^{-\frac{[\bar{y}_1 - f(x_1)]^2}{2s_{\bar{y}_1}^2}} \right) \times \cdots \times \left(\frac{1}{\sqrt{2\pi s_{\bar{y}_M}^2}} e^{-\frac{[\bar{y}_M - f(x_M)]^2}{2s_{\bar{y}_M}^2}} \right) (\Delta y)^M \end{aligned}$$

*This probability is maximized
by choosing parameter values
that minimize the quantity*

$$\chi^2 \equiv \sum_{m=1}^M \frac{[\bar{y}_m - f(x_m)]^2}{s_{\bar{y}_m}^2}$$

Output of a χ^2 / least squares / Gaussian max likelihood fit:

Fit parameters ($a, b, \dots$) that minimize

$$\chi^2 \equiv \sum_{m=1}^M \frac{[\bar{y}_m - f(x_m)]^2}{s_{\bar{y}_m}^2}$$

Parameter uncertainties ($s_a, s_b, \dots$) that tell what shifts in ($a, b, \dots$) would reduce Prob(our results) by a factor of $\sqrt{e}$

Final minimized value of χ^2 and of χ^2 per degree of freedom (also called reduced χ^2): $\# d.f. = M - \# \text{fit params}$

PTE or “probability to exceed”: if the functional form does describe the underlying relationship and the SEMs used were good estimates of error, what is the probability that random errors give you a χ^2 bigger than you got, in a fit with this many degrees of freedom.

This time:

Formalism of weighted χ^2 fitting

Next time (Wed 1/28):

Interpreting a χ^2 fit, plus examples and discussion of how to work with values, residuals, PTE , and special cases.

Lectures Friday 1/30 and forward:

Light as an EM wave, refl/trans at boundaries, interference, diffraction, lensing and more (see lecture plan).

HW1 due Wed 2/4 at 5p.m. on Canvas.